The realization question

OPEN / a quantifier over every finite group

$$\mathrm{Gal}(L/\mathbb{Q}) \cong G$$

C_2	C_3	S_3
$\mathbb{Q}(\sqrt{2})$	$x^3 + x^2 - 2x - 1$	$x^3 - 2$

Examples witness particular groups. They do not settle the universal question.